Suvi Häkkinen, PhD

San Francisco Bay Area • (415) 676 0171 • shakkinen@berkeley.edu • LinkedIn

CURRENT POSITION

Assistant Project Scientist, *University of California, Berkeley* 9/2024 – present

EDUCATION

Doctor of Philosophy (Psychology) 8/2018
University of Helsinki, Finland
Thesis: "Functional organization of human auditory cortex during active auditory tasks"

Master of Science (Technology) 10/2012
Aalto University, Finland
Programme in Bioinformation Technology
Major: Computational and Cognitive Biosciences
Minor: Acoustics and Audio Signal Processing
Thesis: "Temporal dynamics of task-dependent activations of human auditory cortex: an EEG study"

Bachelor of Science (Technology) 8/2012
Aalto University, Finland
Programme in Bioinformation Technology
Major: Computational and Cognitive Biosciences
Minor: Usability in Software Development
Thesis: "Intensity-based registration methods of magnetic resonance images"

RESEARCH EXPERIENCE

Assistant Project Scientist 9/2024 – present
Postdoctoral Scholar 9/2023 – 9/2024
Helen Wills Neuroscience Institute, University of California, Berkeley
Feinberg Lab (PI David Feinberg)

- 7T fMRI quality assessment and analysis to support development of VASO and GRASE sequences for laminar fMRI
- Applying layer-fMRI to neuroscientific research on organization of hippocampal circuits and integration of large-scale functional networks

Postdoctoral Scholar 1/2022 – 9/2023
Helen Wills Neuroscience Institute, University of California, Berkeley
Cognitive Neuroanatomy Lab (PI Kevin Weiner) & Building Blocks of Cognition Laboratory (PI Silvia Bunge)

- Led a project on individual functional-structural variability in tertiary sulci combining manual labeling and machine learning (clustering, classification, regression)
- Coached graduate students on scientific writing, statistics and programming

Postdoctoral Scholar 8/2018 – 12/2021
Memory and Aging Center, University of California, San Francisco (parental leave 5 mo)
Dementia Imaging Genetics Lab (PI Suzee Lee)

- Modeled longitudinal lifespan trajectories in regional volumes, functional networks, fluid biomarkers and cognitive scores, in 3 rare genetic forms of frontotemporal dementia (mixed effects, normative-like modeling, simulation-based cluster correction)
- Collaborated across 17 sites on data quality and subject stratification

- Won a competitive fellowship covering two years research expenses

Researcher 1 – 7/2018
University of Turku, Finland
FinnBrain Birth Cohort Study (PI Hasse Karlsson)

- Developed pipelines to clean and analyze fMRI connectivity acquired in neonates, used in 6 publications and 4 conference abstracts to date

Graduate Researcher 9/2013 – 1/2018
Faculty of Medicine, University of Helsinki, Finland (parental leave 10 mo)
@brain research group (PI Teemu Rinne)

- Managed five research projects on the task-specific activation of human auditory cortex; results published in 3 peer-reviewed journals
- Hypothesis testing based on imaging (fMRI, fMRS, DWI, EEG/MEG) and behavioral data using inferential statistics, clustering, classification, graphs, and time-series analysis
- Tutored graduate researchers and interns in subject recruitment and behavioral training

Research Assistant 3/2008 – 12/2010
Institute of Psychology, University of Helsinki, Finland (part time)
Attention and Memory Networks research group (PIs Kimmo Alho and Teemu Rinne)

- Recruited subjects and scheduled behavioral training and measurement sessions
- 3T MRI scanner operation (Siemens and GE) and data transfer
- Designed and implemented in Python a visualization tool for flattened cortex patches

SKILLS

Programming: Python (NumPy, SciPy, pandas, sklearn, TensorFlow), MATLAB, R; less recently C/C++, Java

Imaging modalities: fMRI (including laminar), DWI, fMRS, EEG/MEG

Imaging software: FreeSurfer, LayNII, FSL, AFNI, SPM, Neuropointillist, fMRIPrep and various other NiPy tools and BIDS apps, MNE, EEGLAB

Misc. technologies: Git, HPC cluster (SGE), SQL, SPSS, LaTeX, Jupyter, shell scripting

Experiments: PsychoPy, Presentation

Languages: Finnish (native), English (fluent), Swedish, German, French

MOOC certificates: Neural Networks and Deep Learning; Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization; Structuring Machine Learning Projects

PUBLICATIONS (* denotes equal contribution/co-first authorship)

Peer-reviewed publications

Vartiainen, E., Copeland, A. Pulli, E., Kumpulainen V., Silver, E. Rajasilta, O., Jolly, A., Luotonen, S., Kholis Audah, H., Bano, W., Suuronen, I., Saukko, E., **Häkkinen (née Talja), S.**, Karlsson, H. , Karlsson, L., Tuulari, J.J. (2025) Pre -and postnatal maternal depressive symptoms associate with local connectivity of the left amygdala in 5-year-olds. *European Psychiatry*, doi:10.1192/j.eurpsy.2025.10097

Flagan, T. M., Chu, S.A., **Häkkinen, S.**, McFall, D., Heller, C., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. (2025) Functional connectivity associations with markers of disease progression in *GRN* mutation carriers. *Annals of Clinical and Translational Neurology,* 12(12):2575–2588.

Häkkinen, S., Voorhies, I.W., Willbrand, E.H., Yi-Heng, T., Gagnant, T., Yao, J.K., Weiner, K.S., Bunge, S.A. (2025) Anchoring functional connectivity to individual sulcal morphology yields insights in a pediatric study of reasoning. *Journal of Neuroscience,* 45(26):e0726242025.

Park, S., Beckett, A., **Häkkinen, S.,** Ma, S.J., Kim, H., Feinberg, D.A. (2025) Higher spatial resolution and sensitivity in whole brain functional MRI at 7T using 3D EPI accelerated with variable density CAIPI sampling and temporal random walk. *Magnetic Resonance in Medicine*, 94(2), 678–693.

Zhang, L., Flagan, T.M., **Häkkinen, S.**, Chu, S.A., Brown, J., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. (2023) Network connectivity alterations in presymptomatic and symptomatic MAPT mutation carriers. *Annals of Neurology*, 94(4), 632–646.

Rajasilta, O., **Häkkinen, S.**, Björnsdotter, M., Scheinin, N. S., ... Tuulari, J.J. (2023) Maternal psychological distress associates with alterations in resting-state low-frequency fluctuations and distal functional connectivity of the neonate medial prefrontal cortex, *European Journal of Neuroscience*, 57, 242–257.

Copeland, A., Korja, R., Nolvi, S., Pulli, E., Kumpulainen, V., Silver, E., Saukko, E., **Häkkinen, S.**, ... Tuulari, J.J. (2022) Maternal sensitivity at the age of 8 months associates with the child local synchrony of the medial prefrontal cortex at 5 years of age. *Frontiers in Neuroscience*, 16.

Rajasilta, O., **Häkkinen, S.**, Björnsdotter, M., Scheinin, N. S., Lehtola, S. J., ... Tuulari, J.J. (2021) Increased maternal BMI during pregnancy is associated with altered neonate brain local synchrony in the left superior frontal gyrus, *Scientific Reports*, 11, 19182.

Häkkinen, S., Chu, S.A., Lee, S.E. (2020) Neuroimaging in genetic frontotemporal dementia and amyotrophic lateral sclerosis. *Neurobiology of Disease*, 145, 205063.

Rajasilta, O., Tuulari, J.J., Björnsdotter, M., Lehtola, S. J., Saunavaara, J., **Häkkinen, S.**, Merisaari, H., Parkkola, R., Lähdesmäki, T., Karlsson, L., Karlsson, H. (2020) Resting state networks of the neonate brain identified in independent component analysis. *Developmental Neurobiology*, 80, 111–125.

Häkkinen, S., Rinne, T. (2018) Intrinsic, stimulus-driven and task-dependent connectivity in human auditory cortex. *Brain Structure & Function*, 223(5), 2113–2127.

Häkkinen, S., Ovaska, N., Rinne, T. (2015) Processing of pitch and location in human auditory cortex during visual and auditory tasks. *Frontiers in Psychology*, 6.

Talja, S., Alho, K., Rinne, T. (2015) Source analysis of event-related potentials during pitch discrimination and pitch memory tasks. *Brain Topography*, 28, 445–458.

Rinne, T., Koistinen, S., **Talja, S.**, Wikman, P., Salonen, O. (2012) Task-dependent activations of human auditory cortex during spatial discrimination and spatial memory tasks. *NeuroImage*, 59, 4126–4131.

Works currently under review

Häkkinen, S., Beckett, A., Walker, E., Huber, R., Feinberg, D.A. Functional imaging of hippocampal layers using VASO on the Next Generation (NexGen) 7T, https://doi.org/10.1101/2025.08.29.673151

Zhang, L.*, **Häkkinen, S.***, Chu, S. A., Flagan, T. M., Coppola, G., Geschwind, D. H., ... Lee, S. E., on behalf of the ARTFL/LEFFTDS/ALLFTD Consortia. Longitudinal functional network connectivity changes across the clinical stages of *C9orf72* hexanucleotide repeat expansion carriers, https://doi.org/10.1101/2025.11.23.25340528

Copeland, A., Rajasilta, O., **Häkkinen, S.**, Kumpulainen, V., Silver, E., Pulli, E.P., Saukko, E., Jolly, A., Saunavaara, J., Parkkola, R., Lähdesmäki, T., Korja, R., Karlsson, L., Karlsson, H., Tuulari, J.J. Functional network organization of the 5-year-old brain identified using independent component analysis and evaluation of denoising strategies, submitted.

CONFERENCE ABSTRACTS

Ru, H., **Häkkinen, S.**, Beckett, A., Walker, E., Feinberg, D.A., Park, S. Accelerated 3D Zoomed GRASE for Mesoscale fMRI using Self-Supervised Reconstruction (DeepGRASE) [Abstract submitted for presentation]. *OHBM,* 2026.

Beckett, A., Walker, E.B., **Häkkinen, S.**, Khegai, O., Vu, A., Huber, R., Feinberg, D. Studying cortical networks using whole brain cerebral blood volume weighted imaging on the NexGen 7T [Abstract submitted for presentation]. *OHBM*, 2026.

Beckett, A.J., **Häkkinen, S.**, Walker, E.B., Khegai, O., Vu, A., Huber, R., & Feinberg, D. (2026, May). Whole-brain, cerebral blood volume weighted imaging of cortical networks using the Next Generation (NexGen) 7T scanner [Abstract submitted for presentation]. *ISMRM*, 2026.

Häkkinen, S., Voorhies, W.I., Willbrand. E.H.,Tsai, Y.-H., Yao. J.K., Weiner, K.S.,, Bunge, S.A. Lateral frontoparietal functional connectivity based on individual sulcal morphology. *SfN*, 2025.

Walker, E. B, Beckett, A. J., **Häkkinen, S.**, Vu, A., Huber, R., & Feinberg, D. Whole-brain, cerebral blood volume weighted imaging of cortical networks using the Next Generation (NexGen) 7T scanner. *SfN*, 2025

Zhang, L., **Häkkinen**, **S.**, Jung, Y., Mandelli, M.L., Leichter, D., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS/ALLFTD Consortia.. *AAIC*, 2025. Longitudinal functional connectivity changes across the clinical spectrum of *C9orf72* hexanucleotide repeat expansion carriers. *AAIC*, 2025.

Häkkinen, S., Beckett, A., Walker, E., Huber, R., Feinberg, D.A. Functional imaging of hippocampal layers using VASO on the Next Generation (NexGen) 7T. *ISMRM,* 2025.

Beckett, A.*, **Häkkinen, S.***, Walker, E., Huber, R., Feinberg, D.A. Functional imaging of hippocampal layers using VASO on the Next Generation (NexGen) 7T. *ISMRM* Joint Workshop of the Ultra-High Field MR & Brain Function Study Groups, 2025. (**Best Poster Award**)

Beckett, A., Park, S., **Häkkinen, S.**, Walker, E., Vu, A., Feinberg, D.A. Development of GRASE Pulse Sequence with larger field of view for mesoscale functional MRI on the Next Generation (NexGen) 7T scanner. *ISMRM,* 2025.

Ma, S.J., Walker, E., **Häkkinen, S.**, Beckett, A.J.S., Vu, A.T., Feinberg, D.A. NexGen 7T MRI scanner for mesoscale brain imaging. *ANA,* 2024.

Flagan, T.M., Chu, S.A., **Häkkinen, S**., Zhang, L., ... Seeley. W.W., Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Functional connectivity associations with markers of disease progression in *GRN* mutation carriers. *AAIC*, 2024.

Huber, R., Beckett, A., Ma, S., Stirnberg, R., Morgan, T., Ehses, P., **Häkkinen, S.,** Gunamony, S., Bandettini, P., Feinberg, S. Developing whole brain layer-fMRI protocols on the NexGen 7T scanner. *OHBM,* 2023.

Beckett, A.J.S., Huber, R., Ma, S. J., **Häkkinen, S.,** Gunamony, S., Feinberg, D.A. Whole brain layer-fMRI on the NexGen 7T scanner with high performance gradients and 64-channel receiver array. *ISMRM,* 2023.

Park, S., Beckett, A.J.S., **Häkkinen, S.,** Ma, S. J., Feinberg, D.A. Whole-brain sub-millimeter resolution fMRI with 3D EPI accelerated with temporal random walk. *ISMRM,* 2023.

Lombardi, J., Zhang, L., Flagan, T.M., Vargas, A.G., Mandelli, M.L., Brown, J.A., **Häkkinen, S.**, ... Miller, B.L., Gorno-Tempini, M.L., Seeley, W.W., Lee, S.E. on behalf of the ALLFTD Consortium. Structural and functional alterations in young adult presymptomatic frontotemporal lobar degeneration mutation carriers. *AAIC*, 2023.

Zhang, L., Flagan, T.M., Staffaroni, A., **Häkkinen, S.**, Brown, J.A., Mandelli, M.L., ... Gorno-Tempini, M.L., Miller, B.L., Seeley, W.W., Lee, S.E. on behalf of the ARTFL/LEFFTDS/ALLFTD Consortia. Functional connectivity trajectories in genetic frontotemporal dementia. *AAIC*, 2023.

Lombardi., J., Zhang, L., Flagan, T. M., Mandelli, M. L., Brown, J. A., **Häkkinen, S.**, ... Lee, S.E., on behalf of the ALLFTD Consortium. Salience network alterations in young adult presymptomatic FTLD mutation carriers. *ISFTD,* 2022.

Copeland, A., Korja, R., Nolvi, S., Pulli, E., Kumpulainen, V., Silver, E., Saukko, E., **Häkkinen, S.**, ... Tuulari, J.J. Maternal sensitivity at the age of 8 months associates with the child local synchrony of the medial prefrontal cortex at 5 years of age. *Flux Congress* 2022.

Flagan T. M., Chu S.A., **Häkkinen S.**, McFall, D., Heller, C., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Complement and NfL associations with brain structure and functional connectivity alterations in presymptomatic and symptomatic *GRN* mutation carriers. *AAIC 2021.*

Zhang, L., Flagan, T.M., Chu, S.A., **Häkkinen, S.**, Brown, J., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Presymptomatic and symptomatic *MAPT* mutation carriers feature functional connectivity alterations. *AAIC,* 2021.

Häkkinen, S., Chu, S.A., Flagan, T. M., Gendron, T. D., Petrucelli, L., ... Lee, S.E., on behalf of the ARTFL/LEFFTDS Consortia. Associations of poly(GP), NfL and functional network alterations in *C9orf72* expansion carriers. *OHBM,* 2020.

Zhang, L., Flagan, T. M., Chu, S.A., **Häkkinen, S.**, Rojas-Martinez, J., ... Lee, S.E., on behalf of the ALLFTD Consortia. Brain functional connectivity changes in presymptomatic and symptomatic *MAPT* mutation carriers. *OHBM,* 2020.

Tuulari, J.J., **Häkkinen, S.**, Rajasilta, O., Lehtola, S., Merisaari, H., ... Karlsson, H. Maternal prenatal anxiety associates to infant orbitofrontal cortex functional synchrony. *Flux Congress*, 2018.

Rajasilta, O., Tuulari J.J., **Häkkinen, S.**, Lehtola, S., Merisaari, H., ... Karlsson, H. Functional networks in term-born infants – a resting-state functional MRI study from FinnBrain Birth Cohort study. *Flux Congress,* 2018.

Rinne, T., **Talja, S.**, Stecker, G.C. (2015) Processing in auditory cortex depends on the characteristics of the discrimination task: fMRI study. *The 38th MidWinter Meeting of The Association for Research in Otolaryngology,* 2015.

Stecker, G.C., Higgins, N., McLaughlin, S., **Talja, S.**, Rinne, T. (2015) Stimulus- and task-dependent spatial processing in the human auditory cortex. *The 38th MidWinter Meeting of The Association for Research in Otolaryngology,* 2015.

Talja, S., Ovaska, N., Rinne, T. (2014) Feature-specific and task-dependent processing of pitch and location in human auditory cortex. *OHBM,* 2014.

Rinne, T., Ovaska, N., **Talja, S.** (2013) Task-dependent activations of human auditory cortex during pitch and spatial tasks. *The 36th MidWinter Meeting of The Association for Research in Otolaryngology*, 2013.

PERSONAL RESEARCH GRANTS

Päivi and Sakari Sohlberg Foundation (Sep 2019, 140 000 €), to support my postdoctoral research on genetic dementia biomarkers

Instrumentarium Science Foundation (Jan 2016, 18 000 €), to support my doctoral research in auditory neuroscience

Fully funded doctoral student position (2014–2016), Doctoral Programme of Psychology, Learning and Communication, University of Helsinki, Finland

Chancellor’s travel grant (2014, 1462 €), University of Helsinki, Finland

TEACHING EXPERIENCE

Presenter at Cognitive Neuroscience Colloquium (brain-wide association studies)	9/2022
Presenter at MAC Postdoc Educational Curriculum (basics of NMR and contrasts)	spring 2019
Assistant supervisor of the Master's thesis of Noora Ovaska	2013

PROFESSIONAL RECOGNITION & SERVICE

Best Poster Award (Joint 1st Author): *ISMRM* Joint Workshop of the Ultra-High Field MR & Brain Function Study Groups, 2025.

Ad-Hoc Reviewer: *Human Brain Mapping*, *Developmental Cognitive Neuroscience*

Public Outreach: Research on tertiary sulci and reasoning (Häkkinen et al., *J. Neurosci*, 2025) featured in *UC Berkeley News*, *Science News Today* and *Neuroscience News*

PROFESSIONAL MEMBERSHIPS

Society of Neuroscience (SfN)	2025
The International Society for Magnetic Resonance in Medicine (ISMRM)	2025
Organization for Human Brain Mapping (OHBM)	2014